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MODULE CODE: CET138

MODULE TITLE: Full Stack Development

MODULE LEADER: David Grey

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ASSIGNMENT: 1

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CET138 Full Stack Development

CSS DOCUMENTATION

Details	
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Live Demo	https://binodkatuwal.github.io/Demos/cssdemo.html

Contents

1. What is CSS?	4
2. Selectors, Specificity and the Box Model.....	4
2.1 Key Selectors	4
2.2 Specificity (who wins when rules conflict)	5
2.3 The Box Model	5
3. Layout – Flexbox and CSS Grid.....	6
3.1 Flexbox (one – dimensional)	6
3.2 CSS Grid (Two – dimensional).....	7
4. Responsive Design and Media Queries	7
5. Transitions, Animations and CSS Variables	8
5.1 Transitions and Transforms.....	8
5.2 CSS Animations	9
5.3 CSS Custom Properties (Variables)	9
6. Summary	10

1. What is CSS?

The language used for controlling the visual presentation of HTML documents is called CSS (Cascading Style Sheets). HTML is used to create structure and content, whereas the colours, fonts, spacing, layout and responsiveness are controlled by CSS. Cascading is the term that is used to describe the set of rules that decide which of multiple rules will apply to the same element, following the rules of specificity, source order and inheritance.

The current standard is CSS3 which consists of different modules like Flexbox, Grid, Animations and Media Queries. External stylesheets (in separate .css files) is the best practice, they separate presentation from content, they can be shared between pages and they are cached by the browser for performance.

2. Selectors, Specificity and the Box Model

```
p      { color: #333; }      /* Type - all <p> elements      */
.card  { padding: 1rem; }   /* Class - class="card"      */
#hero  { background: green; } /* ID - id="hero" (unique per page) */
a:hover { color: darkgreen; } /* Pseudo-class - on mouse hover */
h2::before { content: '→ '; } /* Pseudo-element - inserted content*/
nav a   { font-weight: bold; } /* Descendant combinator      */
ul > li { list-style: square; } /* Direct child combinator     */
input[type="email"] { border: 2px solid green; } /* Attribute selector          */
```

2.1 Key Selectors

2.2 Specificity (who wins when rules conflict)

Selector Type	Specificity	Example
Inline style	1,0,0,0 (highest)	style="color:red"
ID selector	0,1,0,0	#header { }
Class / pseudo-class	0,0,1,0	.nav a:hover { }
Type / pseudo-element	0,0,0,1	p::first-line { }
!important	Overrides all — avoid	color: red !important

2.3 The Box Model

Each HTML element is a four-sided box composed of content, padding, border and margin.

When box-sizing: border-box is used throughout (declared on the root element), padding and border will add to the width specified, causing layout to be much more predictable.

```
*, *::before, *::after { box-sizing: border-box; } /* global reset */

.card {
  width: 320px;           /* total width INCLUDING padding & border */
  padding: 1.5rem;        /* inner space between content and border */
  border: 2px solid #16A34A;
  border-radius: 8px;     /* rounded corners */
  margin: 1rem auto;      /* outer spacing; auto = centre horizontally */
  box-shadow: 0 4px 12px rgba(0,0,0,0.1);
}
```

3. Layout – Flexbox and CSS Grid

3.1 Flexbox (one – dimensional)

Flexbox distributes space along a single axis (row or column). It is ideal for navigation bars, centering content, and card rows.

```
.navbar {  
  display: flex;  
  justify-content: space-between; /* spread items along main axis */  
  align-items: center;           /* align items on the cross axis */  
  gap: 1rem;  
  padding: 1rem 2rem;  
}  
  
/* Equal-width cards that wrap onto new rows */  
.card-row {  
  display: flex;  
  flex-wrap: wrap;  
  gap: 1.5rem;  
}  
  
.card { flex: 1 1 280px; max-width: 360px; } /* grow | shrink | base-size */  
  
/* Perfect centring */  
.hero {  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  min-height: 100vh;  
}
```

3.2 CSS Grid (Two – dimensional)

CSS Grid controls both rows and columns at the same time. It is ideal for full page layouts, dashboards, and galleries..

```
/* Named area page layout */
.page {
  display: grid;
  grid-template-areas:
    'header header'
    'sidebar main '
    'footer footer';
  grid-template-columns: 240px 1fr;
  grid-template-rows: auto 1fr auto;
  min-height: 100vh;
}
header { grid-area: header; }
.sidebar { grid-area: sidebar; }
main { grid-area: main; }
footer { grid-area: footer; }

/* Responsive auto-fill card grid – no media queries needed */
.portfolio {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
  gap: 2rem;
}
```

4. Responsive Design and Media Queries

Responsive design ensures a website works across all screen sizes. The mobile-first approach writes base styles for small screens and adds media queries to enhance the layout for larger screens. Relative units (rem{Root EM}, %{Percentage}, vw{Viewport Width}, vh{Viewport Height}) and clamp() allow fluid scaling without fixed breakpoints.

Short Summary of rem, %, vw, vh

- Rem – consistent scaling based on root font size.
- % – Relative to parent container.
- Vw – scales with screen width
- Vh – scales with screen height..


```

/* Mobile-first base */
.container {
  display: grid;
  grid-template-columns: 1fr; /* single column on mobile */
  padding: 1rem;
}

@media (min-width: 768px) { /* tablet */
  .container { grid-template-columns: 1fr 1fr; padding: 2rem; }
}

@media (min-width: 1024px) { /* desktop */
  .container { grid-template-columns: repeat(3, 1fr); max-width: 1200px; margin: 0 auto; }
}

/* Fluid typography – no breakpoints needed */
h1 { font-size: clamp(1.5rem, 5vw, 3rem); }

/* Dark mode support */
@media (prefers-color-scheme: dark) {
  body { background: #1a1a1a; color: #f5f5f5; }
}

```

5. Transitions, Animations and CSS Variables

5.1 Transitions and Transforms

```

.btn {
  background: #16A34A;
  color: white;
  padding: 0.75rem 1.5rem;
  border-radius: 8px;
  transition: background 0.3s ease, transform 0.2s ease;
}

.btn:hover {
  background: #14532D;
  transform: translateY(-2px); /* lift on hover */
}

.btn:active { transform: translateY(0); } /* press effect */

```


5.2 CSS Animations

```
@keyframes fadeInUp {  
  from { opacity: 0; transform: translateY(24px); }  
  to   { opacity: 1; transform: translateY(0); }  
}  
.hero-title { animation: fadeInUp 0.6s ease-out both; }  
  
@keyframes spin {  
  to { transform: rotate(360deg); }  
}  
.spinner { animation: spin 1s linear infinite; }
```

5.3 CSS Custom Properties (Variables)

```
:root {  
  --color-primary: #16A34A;  
  --color-dark:   #14532D;  
  --radius:       8px;  
  --spacing-md:    1.5rem;  
}  
  
.card {  
  border: 2px solid var(--color-primary);  
  border-radius: var(--radius);  
  padding: var(--spacing-md);  
}  
  
/* Change the whole theme by editing :root only */
```

6. Summary

CSS is the presentation layer of every website. The skills demonstrated in this section cover all core aspects needed by a full stack developer working on the front end:

- Selectors, specificity and the cascade — targeting elements precisely and resolving conflicts
- The box model — understanding how all elements are sized and spaced
- Flexbox — one-dimensional alignment and distribution of items
- CSS Grid — two-dimensional page and component layouts
- Responsive design — mobile-first media queries, relative units, and clamp()
- Transitions and animations — smooth interactive and entrance effects
- CSS Custom Properties — maintainable, theme-ready stylesheets